Central charge windows for the memory calculation Uniform rank tails and a conserved filter correction

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Abstract

We extend the positive rank-product sixth-moment estimate from bounded charge shifts to compact windows of width $\sqrt{N}$. The leading six moments are unchanged. In contrast, replacing the sector variance by its half-filled value inside a growing counterterm leaves a finite error. The exact two-charge memory filter has a further finite, charge-dependent H^2 correction. These results supply inputs for the central-window operator calculation; the full memory Gram matrix in that window is still to be established.

1 The charge window and the literature interface

Use standard CAR, the full overlapping complex Gaussian p-body ensemble, normalized trace at charge m, annealed Hilbert–Schmidt norm, and $\hbar = 1$. The coupling variance w_N is the same in all charge sectors of a given sample. Put

$$\lambda_N = p^2/N, \quad m = N/2 + xp = N/2 + z\sqrt{N}, \quad z = x\sqrt{\lambda_N}. \tag{1}$$

All uniform statements have $|x| \le L$ (equivalently bounded z), and λ_N in a compact subset of $(0, \infty)$. Integer charge values are understood. Limits below keep $\lambda_N = \lambda$; using actual λ_N gives the corresponding uniform expansions.

The fixed-charge embedded-ensemble formulation already supplies the sector variance and reduced spectral parameter [1]:

$$C_m = w_N \binom{m}{p} \binom{N-m+p}{p}, \qquad q_m = \exp\left[-\frac{p^2 N}{m(N-m)}\right]. \tag{2}$$

The expression for C_m is exact; q_m is the double-scaled spectral parameter, not an exact finite-N fourth moment. Fixed-charge and grand-canonical transfer matrices are also treated in Ref. [2]. We use their separation of charge sectors as context; no transfer-matrix formula substitutes for a uniform bound on the marked memory insertions. In particular we keep the common w_N, rather than independently renormalizing the Hamiltonian in each sector.

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2 Uniform sixth moments on a central window

Let U, V be independent isotropic Hermitian rank-a, b matrices, normalized by $\mathbb{E}U^2 = \mathbb{E}V^2 = I$, in charge M. Set

$$\pi_{ab}^{(M)}(r) = \mathbb{E}\,\mathrm{tr}_M[P_r(UV)]^\dagger P_r(UV), \qquad X = x_{a+b} - x_r, \quad x_j = j(N+1-j). \tag{3}$$

The existing positive product law is normalized, has $r \le a+b$, and obeys the Racah product identity [3]:

$$\sum_r \pi_{ab}^{(M)}(r)\chi_r(n) = \chi_a(n)\chi_b(n), \tag{4}$$

$$\chi_a(n) = {}_4F_3\left[\begin{matrix}-a, a-N-1, -n, n-N-1 \\ -M, M-N, -N-1\end{matrix}; 1\right]. \tag{5}$$

Write $P_j(t) = \prod_{\ell<j}(t - x_\ell)$ and

$$c_{nj} = \frac{(-n)_j (n-N-1)_j}{(-M)_j (M-N)_j (-N-1)_j j!}. \tag{6}$$

Then $F_j = \mathbb{E}_\pi P_j(x_r)$ satisfies the exact triangular recursion

$$F_0 = 1, \qquad F_n = \frac{\chi_a(n)\chi_b(n) - \sum_{j<n} c_{nj} F_j}{c_{nn}}. \tag{7}$$

Set $\epsilon = N^{-1/2}$, $a = A/\epsilon$, $b = B/\epsilon$, and $M = 1/(2\epsilon^2) + z/\epsilon$. In the scaled variable $Y = \epsilon^3 x_r$, the coefficient $c_{nj}\epsilon^{-3j}$ is

$$(-4)^j \binom{n}{j} \epsilon^j \prod_{k=0}^{j-1} \frac{1 - (n-1+k)\epsilon^2}{(1 + 2z\epsilon - 2k\epsilon^2)(1 - 2z\epsilon - 2k\epsilon^2)(1 + (1-k)\epsilon^2)}. \tag{8}$$

This is the substantive change from a bounded shift $M - N/2$. The roots are $\epsilon^3 x_\ell = \ell\epsilon + \ell(1-\ell)\epsilon^3$ and the input is $Y_a = A - A^2\epsilon + A\epsilon^2$.

The exact series recursion over $\mathbb{Q}[A, B, z]$ through ϵ^{12} cancels all 21 negative Laurent coefficients in the first six moments of X/N. Its constant terms give

$$\mathbb{E}_\pi (X/N)^j = T_j(2AB) + O(\epsilon), \quad 1 \le j \le 6, \qquad T_{j+1}(t) = t[T_j(t) + T_j'(t)],\ T_0 = 1. \tag{9}$$

After extracting the known powers of ϵ, each divisor in (7) has nonzero constant term $(-4)^n$. All other denominators in (8) remain nonzero on compact A, B, z sets for sufficiently small ϵ. Consequently the exact rational functions are analytic there and the remainders in (9) are uniform, not merely formal.

3 What the tail bound controls

The sixth polynomial is

$$T_6(t) = t^6 + 15t^5 + 65t^4 + 90t^3 + 31t^2 + t. \tag{10}$$

For $a, b \le p$, the output deficit $S = a + b - r$ is nonnegative and

$$X = S[N + 1 - 2(a + b) + S] \ge S(N + 1 - 4p). \tag{11}$$

Thus, uniformly on the central window and on its adjacent charges,

$$\sup_{a,b\le p,\ |M-N/2|\le L\sqrt{N}} \mathbb{E}_\pi S^6 \le K_{L,\lambda}, \qquad \sum_{S>R} \pi_{ab}^{(M)}(r) \le K_{L,\lambda} R^{-6}. \tag{12}$$

The norm of an independent rank-product tail is the square root of this probability. The constant is allowed to depend on L; the calculation does not cover a growing $L = L_N$.

This distribution refers to independent isotropic rank inputs. The actual memory operator contains coherent products in adjacent charges and ordinary Gaussian chaos subtractions. Applying (12) to that operator still requires the appropriate charged norm-transfer and input-rank estimates. It is not a direct bound on the whole memory remainder.

Executed checks. The polynomial-ring calculation verifies six constant terms and 21 cancellations. The four full finite distributions at

$$(N, M, a, b) = (16, 6, 2, 3), (16, 10, 2, 3), (20, 7, 3, 3), (20, 13, 3, 3)$$

independently reproduce all 28 moments from order zero to six. Their S^6 bounds are checked as exact fractions. Forty further exact-rational cases use $\lambda = 1/4$, $p = 16, 32, 64, 128$, $z = -2, -1, 0, 1, 2$, and input deficits $(0, 0), (1, 2)$; reflection $M \leftrightarrow N - M$ is checked explicitly.

For example, at $p = 128$, $z = 2$, $a = b = p$, $\mathbb{E}(X/N)^6 = 22.4617786805$, while $T_6(1/2) = 24.046875$. The finite difference is retained; these are convergence checks, not equality to the limiting moments at finite N.

4 A vanishing variance difference leaves a finite counterterm

Let $C_0 = C_{N/2}$ for even N. An exact product from (2) is

$$\frac{C_m}{C_0} = \prod_{i=0}^{p-1} \left(1 + \frac{xp}{N/2 - i}\right) \left(1 - \frac{xp}{N/2 + p - i}\right). \tag{13}$$

With $\eta = 1/p$ and $y = i/p$, the logarithm of each paired factor has the uniform expansion

$$\begin{aligned} \log\left(1 + \frac{2\lambda x\eta}{1 - 2\lambda y\eta}\right) + \log\left(1 - \frac{2\lambda x\eta}{1 + 2\lambda(1-y)\eta}\right) \\ = 4\lambda^2 x(1-x)\eta^2 + O(\eta^3). \end{aligned} \tag{14}$$

Summing p terms and exponentiating gives

$$\boxed{p\left(\frac{C_m}{C_0} - 1\right) = 4\lambda^2 x(1-x) + O(p^{-1}).} \tag{15}$$

The asymmetric center is not a mistake: the exact variance obeys $C_m = C_{N+p-m}$, corresponding to $x \mapsto 1 - x$. It is maximal near $m = (N+p)/2$, not exactly at half filling. The original ensemble contains all overlapping index sets; no particle-hole normal-ordering modification has been made.

Meanwhile

$$\log q_m = -\frac{4\lambda}{1 - 4\lambda^2 x^2/p^2} = -4\lambda - \frac{16\lambda^3 x^2}{p^2} + O(p^{-4}). \tag{16}$$

Thus a common leading spectral shape can coexist with a finite correction to a subtraction proportional to pC_m. Replacing C_m by C_0 in that subtraction is not justified by $C_m/C_0 \to 1$.

Thirty exact binomial ratios, $p = 16, 32, 64, 128, 256$ and $x = -2, -1, 0, 1/2, 1, 2$, verify the reflection identity and record the deviation from (15). These use a common coupling variance; no independent sector rescaling is performed.

5 The raw core filter also acquires a finite charge correction

Use the exact previous two-charge decomposition [3], with $A_{ab;r} = Z^{-}_{ab;r} + (2p-m)Z^{+}_{ab;r}$:

$$Q_{\mathcal{E}} = \sum_{a\le b,r} (g_H Z^{+}_{ab;r} + g_A A_{ab;r}). \tag{17}$$

Here $Q_{\mathcal{E}}$ is the quadratic contraction of the raw quartic core, prior to its scalar centering. Its known rational coefficients can be organized as

$$g_H/C_m = B_0 + B_a\kappa_a/C_m + B_b\kappa_b/C_m + B_r\kappa_r/C_m, \quad \kappa_j = w_N \binom{m-j}{p}\binom{N-m+p-j}{p}. \tag{18}$$

For $a = p-u$, $b = p-v$, $r = 2p-u-v-S$, fixed deficits, and $m = N/2 + xp$, the exact rational expansions are

$$B_0 = 2p + \lambda + 4\lambda x + O(p^{-1}), \tag{19}$$

$$B_a = B_b = -2p - 5\lambda - 4\lambda x + O(p^{-1}), \tag{20}$$

$$B_r = 2p + 9\lambda + 4\lambda x + O(p^{-1}). \tag{21}$$

For $j = tp - s$, the paired binomial product gives

$$\log(\kappa_j/C_m) = -4\lambda t + (4\lambda s - 4\lambda^2 t^2)/p + O(p^{-2}). \tag{22}$$

The charge coordinate cancels through this order. Therefore, with $q = e^{-4\lambda}$,

$$\begin{aligned} g_H/C_m &= 2p(1-q)^2 + K_0 + 4\lambda x(1-q)^2 \\ &\quad + 8\lambda(q^2-q)(u+v) + 8\lambda q^2 S + O(p^{-1}), \end{aligned} \tag{23}$$

$$K_0 = \lambda[1 - 10q + 9q^2 + 16\lambda q - 32\lambda q^2]. \tag{24}$$

The additional term is rank-independent. After ordinary Gaussian centering it multiplies $C_m(h_m^2 - C_m I)$, a conserved direction. It must be included before studying the remaining dynamic field.

For g_A/C_m the four rational prefactors, at arbitrary bounded rank proportions, remain

$$(3/2, -1, -1, 1/2) + \frac{\lambda}{p}(8, -5/2, -5/2, 3) + O(p^{-2}). \tag{25}$$

In particular the fixed-deficit leading coefficient is still $(3-4q+q^2)/2$. Uniform operator convergence additionally needs the coherent norm estimates identified on page 3.

6 Verification and the next operator calculation

The saved finite five-word implementation is independent of the rational-coefficient expansion. At $\lambda = 1/4$, five values $p = 16, 32, 64, 128, 256$, $x = -1, 0, 1$, and two deficit triples, it verifies 60 exact rational identities for g_H and g_A. The normalized difference

$$g_H(N/2 + xp)/C_m - g_H(N/2)/C_0 \longrightarrow 4\lambda x(1-q)^2 \tag{26}$$

is evaluated at 80 and 120 decimal digits with agreement better than 10^{-70} in the stored quantities. For $p = 256$, $x = 1$, $(u, v, S) = (2, 1, 3)$, its value is 0.39841612746439; the limiting value is $(1 - e^{-1})^2 = 0.39957640089373$. The finite discrepancy is recorded, not removed by fitting. Zero-limit cases use absolute errors. These are exact-rational inputs followed by high-precision evaluations, not interval arithmetic.

Reproducibility. The three records are `central-moments.json`, `central-scale.json`, and `central-filter-checks.json` under `results/susskind-memory-central-window`. They retain parameter choices, dependency versions, and input hashes. The first symbolic run exposed a reused dummy variable; the first scale check used structural rather than algebraic symbolic equality. Both initial sources and explanations are preserved. Corrections kept exact algebraic assertions. Previous scientific records were not overwritten.

Continuation. The central-window tail estimate and the finite conserved correction are now explicit. Next derive the actual charged norm transfer and marked overlap estimates uniformly on this window, including the A input-rank tail, then assemble the memory Gram matrix with sector-dependent conserved subtractions. Compact-window control by itself does not justify summing the whole charge distribution; an integrable bound outside the window is also required. After that, control the two-insertion finite-memory remainder and connect the microscopic fields to the Susskind singlet/meson spatial and mass dictionary.

References

[1] J. Tall and S. Tomsovic, arXiv:2604.14522v2, Secs. 2.1 and 6: fixed-sector normalization and reduced moments.

[2] S. Förste, Y. Kruse and S. Natu, arXiv:2512.07715v2, Secs. 3.1–3.2: canonical and grand-canonical transfer matrices.

[3] A. Day and Chengheng, local notes: *An exact two-charge formula for the memory core*; *The quadratic memory core in the double-scaled limit*; *Restoring the dynamic memory descendants* (2026-11-06).